

## 1 a.e. integrability

**Theorem 1.1.** Suppose  $E \in \mathcal{F}$  is such that  $\mu(E^c) = 0$  then,

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu,$$

provided either side exists. (In fact, if one side exists, then so does the other, and they are equal)

**Proof. Step 1:** Let  $f \geq 0$  be measurable and suppose  $\int_{\Omega} f \, d\mu < \infty$ , Then

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu + \int_{E^c} f \, d\mu.$$

But

$$\int_{E^c} f \, d\mu = \int_{\Omega} f \mathbf{1}_{E^c} \, d\mu,$$

and  $0 \leq f \mathbf{1}_{E^c} \leq \infty \cdot \mathbf{1}_{E^c}$ , which by infinity arithmetic gives,

$$0 \leq \int_{\Omega} f \mathbf{1}_{E^c} \, d\mu \leq \infty \cdot \mu(E^c) = 0.$$

Hence  $\int_{E^c} f \, d\mu = 0$ , therefore

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu.$$

**Step 2:** For general  $f$ , write  $f = f^+ - f^-$ . Suppose first that

$$\int_{\Omega} f^+ \, d\mu < \infty.$$

Then, by Step 1,

$$\int_{\Omega} f^+ \, d\mu = \int_E f^+ \, d\mu.$$

Hence

$$\begin{aligned} \int_{\Omega} f \, d\mu &= \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu \\ &= \int_E f^+ \, d\mu - \int_E f^- \, d\mu \\ &= \int_E f \, d\mu. \end{aligned}$$

Now suppose  $\int_{\Omega} f^- \, d\mu < \infty$ , then the same argument gives,

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu.$$

Finally, if  $\int_{\Omega} f^+ \, d\mu = \infty$  then,

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu = +\infty$$

Whenever  $\int_{\Omega} f^- \, d\mu < \infty$ , by Step 1,

$$\int_E f^+ \, d\mu = \infty, \quad \int_E f^- \, d\mu < \infty,$$

and hence

$$\int_E f \, d\mu = +\infty$$

The case for  $\int_\Omega f^- \, d\mu = \infty$  is analogous.  $\square$

## 2 Change of Variables Formula

Suppose  $(\Omega, \mathcal{F}, \mu) \xrightarrow{f} (\Omega', \mathcal{F}', \nu)$  and  $\nu = \mu \circ f^{-1}$  where  $f$  is measurable. We want to write

$$\int_{\Omega'} g(\omega') \, d\nu(\omega')$$

in terms of the measure space  $(\Omega, \mathcal{F}, \mu)$

## 3 Change of Variables Formula

**Theorem 3.1 (Change of Variables Formula).** Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $(\Omega', \mathcal{F}')$  be a measurable space. Let  $f : \Omega \rightarrow \Omega'$  be  $\mathcal{F}/\mathcal{F}'$ -measurable. Let  $\nu = \mu \circ f^{-1}$ .

Then, for every non-negative measurable function  $g : \Omega' \rightarrow [0, \infty]$ ,

$$\int_B g \, d\nu = \int_{f^{-1}(B)} g \circ f \, d\mu \quad (*) \quad \forall B \in \mathcal{F}'.$$

The same identity holds for every integrable measurable  $g$ .

**Corollary 3.2.** Suppose  $f : \Omega \rightarrow \mathbb{R}$  is measurable, and let  $\nu = \mu \circ f^{-1}$ . Then

$$\int_\Omega f \, d\mu = \int_{\mathbb{R}} x \, \nu(dx).$$

Thus the integral of  $f$  can be written as an integral with respect to the distribution induced by  $f$ .

**Proof.** We prove the result by measure-theoretic induction on  $g$ .

**Step 1:** Suppose  $g = \mathbf{1}_C$ , and  $C \in \mathcal{F}'$ . Then,

$$\begin{aligned} \int_B g \, d\nu &= \nu(C \cap B) \\ &= \mu(f^{-1}(C \cap B)) \\ &= \mu(f^{-1}(C) \cap f^{-1}(B)) \\ &= \int_{f^{-1}(B)} \mathbf{1}_{f^{-1}(C)} \, d\mu = \int_{f^{-1}(B)} \mathbf{1}_C \circ f \, d\mu. \end{aligned}$$

Hence

$$\int_B \mathbf{1}_C \, d\nu = \int_{f^{-1}(B)} \mathbf{1}_C \circ f \, d\mu.$$

**Step 2:** If the result holds for  $g_1, \dots, g_N \geq 0$  and  $\alpha_1, \dots, \alpha_N > 0$ , then, by linearity,  $g = \sum_{i=1}^N \alpha_i g_i$  also satisfies  $(*)$ . Hence  $(*)$  holds for every non-negative simple measurable function.

**Step 3:** Suppose  $(*)$  holds for  $g_n \geq 0$  and  $g_n \uparrow g$ . Then, by the Monotone Convergence Theorem,

$$\begin{aligned} \int_B g \, d\nu &= \lim_{n \rightarrow \infty} \int_B g_n \, d\nu \\ &= \lim_{n \rightarrow \infty} \int_{f^{-1}(B)} g_n \circ f \, d\mu \\ &= \int_{f^{-1}(B)} g \circ f \, d\mu. \end{aligned}$$

Since every non-negative measurable function is the pointwise limit of an increasing sequence of non-negative simple functions,  $(*)$  holds for every  $g \geq 0$  measurable.

**Step 4:** For a general measurable  $g$ , write  $g = g^+ - g^-$ . Applying the result separately to  $g^+$  and  $g^-$ , whenever the corresponding integrals are finite, gives

$$\int_B g \, d\nu = \int_{f^{-1}(B)} g \circ f \, d\mu.$$

Hence the result follows.  $\square$

## 4 Riemann Integral

Let  $-\infty < a < b < \infty$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Let  $P = \{x_0, x_1, \dots, x_n\}$  be a partition of  $[a, b]$ , i.e.  $a = x_0 < x_1 < \dots < x_n = b$ .

Define the norm (mesh) of  $P$  by

$$\|P\| = \max_{0 \leq i \leq n-1} (x_{i+1} - x_i).$$

We say  $P'$  is finer than  $P$  if  $P \subseteq P'$ . Then  $\|P\| \geq \|P'\|$ .

Define the *Riemann upper sum* as

$$U(f, P) = \sum_{i=0}^{n-1} \sup_{x \in [x_i, x_{i+1}]} f(x) (x_{i+1} - x_i),$$

and the *Riemann lower sum* as

$$L(f, P) = \sum_{i=0}^{n-1} \inf_{x \in [x_i, x_{i+1}]} f(x) (x_{i+1} - x_i).$$

By definition,  $U(f, P) \geq L(f, P)$ . If  $P'$  is finer than  $P$ , then

$$L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P).$$

Let  $\{P_n\}_{n=1}^\infty$  be a sequence of partitions such that

$$P_n \subseteq P_{n+1} \quad \text{and} \quad \|P_n\| \rightarrow 0.$$

Then

$$L(f, P_n) \uparrow \int_a^b f(t) \, dt,$$

where the limit is called the *Riemann lower integral*, and

$$U(f, P_n) \downarrow \overline{\int_a^b f(t) \, dt},$$

where the limit is called the *Riemann upper integral*.

**Remark 4.1.** If

$$Q_n \subseteq Q_{n+1}, \quad \|Q_n\| \rightarrow 0$$

is another sequence of partitions, then both the lower and upper integrals obtained above are the same. Indeed, one may take common refinements of  $P_n$  and  $Q_n$ , whose mesh also tends to 0. Thus the Riemann lower and upper integrals are independent of the particular nested sequence of partitions used.

**Definition 4.2.** A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is called *Riemann integrable* iff

$$\int_a^b f(t) \, dt = \overline{\int_a^b f(t) \, dt}.$$

Their common value is called the *Riemann integral* of  $f$ , and is denoted by

$$\Re \int_a^b f(t) \, dt.$$

**Note.** In these set of notes, for the sake of clarity we write Riemann integrals as  $\Re \int$  and Lebesgue integrals as  $\int$

**Example 4.3.** Consider  $f = \mathbf{1}_{\mathbb{Q} \cap [a,b]}$ . Then every subinterval contains both rational and irrational numbers, and hence

$$\int_a^b f(t) dt = 0, \quad \overline{\int_a^b} f(t) dt = b - a.$$

Therefore  $f$  is not Riemann integrable.

**Remark 4.4.** Consider  $([a, b], \mathcal{B}_{[a,b]}, \lambda)$  where  $\lambda$  denotes Lebesgue measure. For a partition  $P_n = \{x_0, \dots, x_n\}$  define the step functions

$$U_n(f, x) = \begin{cases} \sup_{x_i \leq x \leq x_{i+1}} f(x), & x_i < x \leq x_{i+1}, \\ 0, & x = a, \end{cases}$$

$$L_n(f, x) = \begin{cases} \inf_{x_i \leq x \leq x_{i+1}} f(x), & x_i < x \leq x_{i+1}, \\ 0, & x = a. \end{cases}$$

Then

$$U(f, P_n) = \int_{[a,b]} U_n d\lambda, \quad L(f, P_n) = \int_{[a,b]} L_n d\lambda.$$

Moreover,  $U_n(x) \downarrow U(x)$  and  $L_n(x) \uparrow L(x)$  and both  $U_n, L_n, U, L$  are measurable.

Hence, by the Monotone Convergence Theorem,

$$\int_{[a,b]} L_n d\lambda \uparrow \int_{[a,b]} L d\lambda, \quad \text{and} \quad \int_{[a,b]} U_n d\lambda \downarrow \int_{[a,b]} U d\lambda.$$

If  $f$  is Riemann integrable, then

$$R \int_a^b f(t) dt = \int_a^b f(t) dt = \overline{\int_a^b} f(t) dt.$$

Therefore

$$\int_{[a,b]} L d\lambda = \int_{[a,b]} U d\lambda = R \int_a^b f(t) dt.$$

Since  $L(x) \leq f(x) \leq U(x)$ , we have

$$\int_{[a,b]} (U - L) d\lambda = 0, \implies \boxed{L = U \text{ a.e. } [\lambda]}$$

Therefore  $f$  is almost surely measurable and

$$\int_{[a,b]} f d\lambda = \Re \int_a^b f(t) dt.$$

**Theorem 4.5 (Lebesgue's Criterion for Riemann Integrability).** Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded.

(a) If  $f$  is Riemann integrable, then  $f$  is almost surely measurable and

$$\int_{[a,b]} f d\lambda = R \int_a^b f(t) dt.$$

(b)  $f$  is Riemann integrable **if and only** if the set of points of discontinuity of  $f$  has Lebesgue measure zero.

**Remark 4.6.** If  $x \notin \bigcup_{n=1}^{\infty} P_n$  then the intervals containing  $x$  shrink to  $x$ . Hence  $L(x) \leq f(x) \leq U(x)$ . In particular, if  $L(x) = U(x)$  then  $f$  is continuous at  $x$ .

## 5 Non-Classical Set-Up

The classical Riemann integral assumes:

1. the domain is a compact interval  $[a, b]$
2.  $f$  is bounded;
3. hence the function and the interval both satisfy the classical assumptions.

We now consider situations where one or both of these assumptions fail.

**Example 5.1.** Let

$$f(x) = \frac{(-1)^{n-1}}{n}, \quad n-1 \leq x \leq n, \quad n \geq 1.$$

Then  $f$  is defined on  $[0, \infty)$  and

$$\Re \int_0^N f(t) dt = \sum_{n=1}^N \frac{(-1)^{n-1}}{n}.$$

Hence

$$\lim_{N \rightarrow \infty} \Re \int_0^N f(t) dt = \ln 2.$$

Thus the improper Riemann integral exists. ( $\Re \int_0^\infty f(t) dt = \ln 2$ ). However,

$$f^+(x) = \frac{1}{2n+1}, \quad f^-(x) = \frac{1}{2n},$$

on the corresponding intervals. Hence

$$\int_0^\infty f^+ d\lambda = \int_0^\infty f^- d\lambda = \infty.$$

Therefore  $f$  is not Lebesgue integrable.

**Remark 5.2.** The improper Riemann integral can exist even though the Lebesgue integral does not exist. The issue is that the positive and negative parts are both infinite.

**Example 5.3.** Similarly,

$$\int_{-\infty}^\infty \frac{\sin x}{x} dx$$

can be considered as an improper Riemann integral, and we obtain,

$$\Re \int_{-\infty}^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$$

under the usual symmetric/improper interpretation.

**Example 5.4.** Consider  $f : \mathbb{R} \rightarrow \mathbb{R}$  with  $f(x) = x/(1+x^2)$ . Then,

$$\lim_{M, N \rightarrow \infty} \Re \int_{-N}^M f(t) dt$$

does not exist.

Indeed,

$$\Re \int_{-N}^M \frac{t}{1+t^2} dt = \frac{1}{2} \log \left( \frac{1+M^2}{1+N^2} \right),$$

whose limit depends on how  $M, N \rightarrow \infty$ .

However the symmetric limit exists as,

$$\lim_{N \rightarrow \infty} \Re \int_{-N}^N f(t) \, dt = 0.$$

This is called the *Cauchy principal value*. This is often useful in physics for example in problems of calculating moments of inertia.

**Example 5.5.** Consider

$$f(x) = \frac{1}{\sqrt{x(1-x)}}, \quad 0 < x < 1.$$

Then  $f$  is unbounded near 0 and 1, so the classical Riemann integral does not apply directly.

Nevertheless, the improper Riemann integral exists, and the corresponding Lebesgue integral also exists.